

Results (Place and Route, Vitis25, xczu7ev-ffvc1156-2-e, 1 PE, all driver+collectors)

	Main loop (cycles)	II (cycles)	Latency (ns)	Frequency (MHz)	LUT	FF	DSP	SRL	CLB
No syscredit	4	1	4.297	233Mhz	4244	3260	2	224	929
With syscredit	4	1	4.372	229Mhz	5095	4108	2	240	1049

Cosim

- 1 PE -> pass for both versions
- 4x4 -> pass with syscredit -> MAC throughput 1/5.3 MAC/cycle

MAC throughput (1 PE)

Original (limit for a single PE=1/4?)	Ours (no syscredit)
1/4 MAC/cycle	1/4 MAC/cycle

* due to reprogrammability -> 1 MAC = 4 instructions

EVA original values (ASIC)

GEMM	567.57 GMACs/s
Frequency	100MHz-1.5GHz

Source: EVA: A 16mm2 1.54TFLOPS Tiled-Based Accelerator for Evolvable Edge Computing

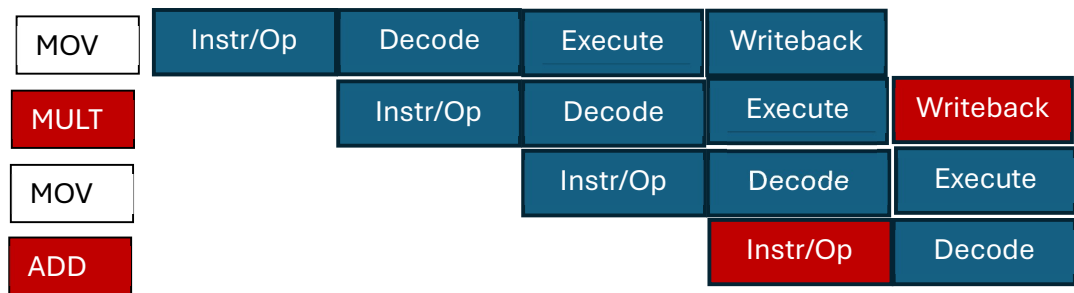
EVA's MAC program – dependency problem

MOV r1, syslft # 0011

MULT r2, r0, r1 # 0010

MOV sysrgt, r1 # 0011 move to the right

ADD sysbtm, r2, systop # 0000 move to the bottom



Instruction 4 needs the result from instruction 2 !!!!